

Reproduction of arXiv:2309.12402 with SDPB

A comparison with the figures of the paper

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1 What was done

These are the results of our attempt to reproduce *Bootstrapping gauge theories* (arXiv:2309.12402, v3), compared figure by figure with the paper.

We set the finite problem of Section 3 up directly as a polynomial matrix program and solved it with SDPB, at high precision; no other solver was involved. The discretised operators (*grid and conformal map, cot kernel, angular projections of the Mandelstam representation, current kernels*) were rederived from Sections 2 and 3 rather than taken from any existing code, and cross-checked with a Mathematica script. Every solution quoted below was re-verified against the original constraints in interval arithmetic (Arb) after the solve.

Table 1 lists the inputs we took as printed. There are five numerical choices the paper does not spell out, and they turn out to matter for Figs. 8–10; Table 2 lists them with the readings we tried.

Table 1: Inputs taken as printed in the paper.

Discretisation	$M = 50$, $\phi_i = (i - \frac{1}{2})\pi/M$, $\nu_0 = 0$; unitarity imposed for $L = 10$ waves per isospin
Matching scale and QCD inputs	$s_0 = (1.2 \text{ GeV})^2$, $\alpha_s = 0.4$, $m_u = 4 \text{ MeV}$, $m_d = 7.3 \text{ MeV}$, the condensates of (2.54), $m_\pi = 140 \text{ MeV}$, $f_\pi = 92 \text{ MeV}$
Sum rules	the printed numbers of (2.56) times s_0^{n+2} ; moments $n = 0, 1$ for S0 and $n = -1, 0$ for P1; $\epsilon^{SR} = 2 \times 10^{-3}$
Form factors	$\epsilon^{FF} = 6 \times 10^{-5}$ in (3.75)
Chiral matching	points $s = \frac{1}{2}, 1, \frac{3}{2}, 2$; $\epsilon^\chi = 2 \times 10^{-3}$

Table 2: Choices the paper does not spell out, and the readings we tried.

Item	Paper text	Readings run
Norm of the sum-rule tolerance (3.73), $\epsilon^{SR} = 2 \times 10^{-3}$	<i>with some norm</i>	a per-moment box on the raw moments (<i>main line</i>); a per-wave L^2 ball (<i>the structure used in the code released with [arXiv:2403.10772]</i>); a relative 10 % box per moment
Evaluation point of the factor between F and $\mathcal{F}$ in (3.75)	<i>which we evaluate at $s = s_0$ (in the estimate of ϵ^{FF})</i>	the factor at each node $s_i > s_0$ (<i>main line</i>); the factor frozen at s_0
Regularisation of the double spectral density	not mentioned in the paper; the earlier method paper [arXiv:2103.11484], Section 3, and the code released with [arXiv:2403.10772] use an M -bound	$ \rho_{ij} \leq M_{\text{reg}}$, with M_{reg} fixed by an in-model rule (<i>omitted-wave unitarity and L-stability</i>): 10^2 for the chiral and UV stages, 10^3 for the pure stage; ℓ^2 , ℓ^4 and $M_{\text{reg}} \times 10$ controls
Amplitude at a boundary point	<i>only points at the boundary have partial waves associated with them</i>	the solver's optimal point, together with a diagnostic of the whole near-optimal face (<i>Section 4</i>)
Chiral norm in (3.64)	<i>with some norm</i>	one combined 8-dimensional L^2 norm (<i>as in the code released with [arXiv:2403.10772]</i>); two separate 4-dimensional norms as a control

2 Summary

Table 3: Summary; details and figures in Section 3.

Figure	Paper statement	Our result	Agreement
Fig. 3	pure-unitarity region, $M = 50, L = 10$	$+x$ end 2.22445 vs 2.23289 (-0.4%); $-x$ end -2.247 vs -2.902 ; f_1^1 range $[-0.6243, 0.06629]$ vs $[-0.7336, 0.07932]$	the $+x$ end agrees; our region lies inside yours on the $-x$ and $\pm y$ sides
Fig. 4	the chiral constraints collapse the region onto $f_1^1 = -f_0^0/15$	$\epsilon^X = 2 \times 10^{-3}$: $+x$ end 0.082757 vs 0.0825728 ($+0.2\%$); x_{ref} section width 7.618×10^{-4} vs 7.598×10^{-4} ; the six-tolerance ladder is monotone	agrees
Fig. 5	the subthreshold waves are nearly linear and the S0 chiral zero moves with ϵ^X	$\text{RMS} \leq 6.4\%$ of $f_0^0(3)$ against the digitised curves at $\epsilon^X = 2, 4, 6 \times 10^{-3}$; S0 zero at $s = 0.426, 0.293$, absent	agrees
Fig. 7	with the chiral constraints alone S0 and S2 agree with experiment and P1 has no ρ	RMS against the digitised curves S0 0.388° , S2 0.144° , P1 0.456° ; no P1 crossing below 1.2 GeV	agrees
Fig. 8	with the sum rules the upper boundary shrinks much more than the lower one	upper shrink / lower rise at x_{ref} : 2.8×10^{-4} / 2.8×10^{-4} (factor at each node), 5.3×10^{-5} / 2.5×10^{-4} (factor frozen at s_0); paper 2.32×10^{-4} / 3.7×10^{-5} . $+x$ end 0.076013 / 0.079623 vs 0.08112	the asymmetry does not appear in any feasible reading
Fig. 9	P1 phase shift with the ρ ; the 90° crossing lies about 6% above 770 MeV	crossings: tip 772.7 MeV ($ S _{\text{min}} = 0.585$), mid 738.9 MeV ($ S _{\text{min}} = 0.077$); at the reference point $ S $ dips to 0.24 at 0.73 GeV (708 MeV on the $+180^\circ$ branch); paper 813–827 MeV	a ρ -like structure appears at every point once the UV constraints are added; it lies 40–110 MeV below yours and is strongly inelastic
Fig. 10	S0 and S2 coincide at low energy and spread above	S2: RMS 0.354–1.44°, endpoints -28 to -32° (paper -18 to -36°); S0: RMS 5.7–16.6°, endpoint at 1.196 GeV 181–192° (paper 87–106°)	S2 agrees; S0 rises faster than yours above 0.6 GeV
Fig. 11	good convergence in L ; the ρ peak shifts with M	P1 crossing at $M = 50, L = 8/10/12$: 771.4 / 772.7 / 772.9 MeV; $L = 10, M = 45/50/60$: 736.1 / 772.7 / 738.8 MeV; S0 at 1 GeV 150–165° (paper about 95°)	the L dependence agrees; the shift with M is 37 MeV; the S0 level differs

3 Figure-by-figure comparison

Left (or top) panels are the figures from the arXiv source; right (or bottom) panels are our solutions with the digitised curves or boundaries overlaid. Quotations from the paper are in *grey italics*.

3.1 Fig. 3, pure unitarity region

“we have used $M = 50$ in (3.61) for discretization and imposed unitarity for 10 partial waves per isospin.”

We ran 24 support directions, 15° apart, at $M = 50$, $L = 10$ and $M_{\text{reg}} = 10^3$ (*the scale at which the first omitted waves stay unitary to 2% and the $L = 8/10/12$ endpoints agree*). Against the digitised figure the extrema are f_0^0 max 2.22445 vs 2.23289 (-0.4%), f_0^0 min -2.247 vs -2.902 , f_1^1 min -0.6243 vs -0.7336 and f_1^1 max 0.06629 vs 0.07932. The $+x$ end hardly moves with M_{reg} (2.2116, 2.2245, 2.2305 at 10^2 , 10^3 , 10^4). The other three extrema come out inside your region by 15–23%; these are the directions where the amplitudes get large and the $|\rho_{ij}|$ bound bites, so we read them as a measure of the regulariser rather than of the model. A run at $M_{\text{reg}} = 10^4$ in these directions is still going.

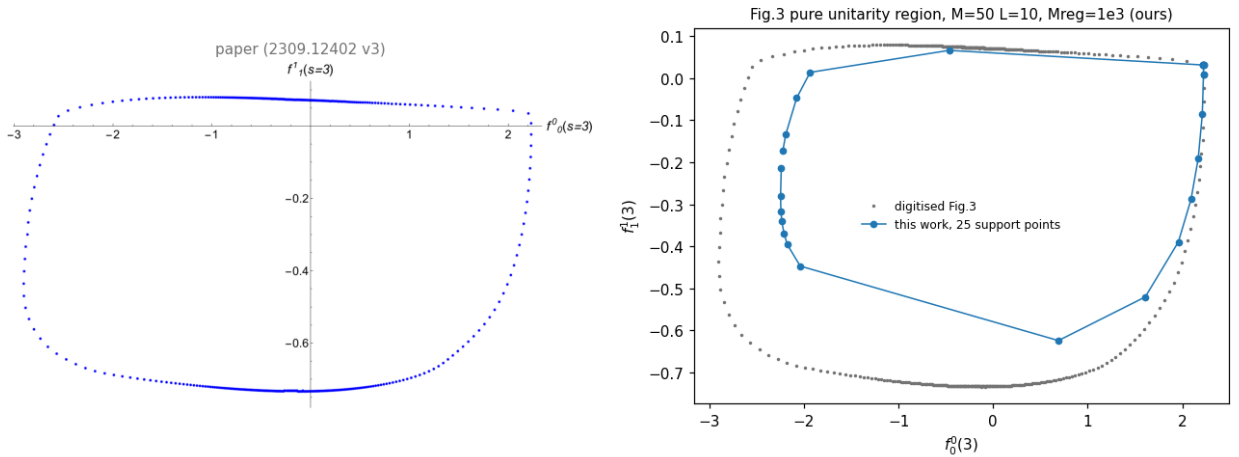


Fig. 3: the paper (left) and our 24 support points with the digitised boundary (right).

3.2 Fig. 4, chiral constraints

“restricted by the chiral constraints (3.64) with tolerances 6×10^{-3} , 4×10^{-3} , 2×10^{-3} , 1×10^{-3} , 6×10^{-4} , 2×10^{-4} (from the outer shape inward) ... with some norm”

We impose the chiral constraints as a single L^2 norm over the eight residuals at the four points, which is what the code released with [arXiv:2403.10772] does, with $M_{\text{reg}} = 10^2$. At $\epsilon^x = 2 \times 10^{-3}$ the $+x$ end is 0.082757 (paper 0.0825728, $+0.2\%$) and the vertical section at x_{ref} has width 7.618×10^{-4} (paper 7.598×10^{-4}). The $+x$ ends for the six tolerances are 0.1608, 0.1255, 0.0828, 0.0551, 0.0411, 0.0223 (from 6×10^{-3} down to 2×10^{-4}). Splitting the norm into two separate ones moves the $+x$ end by less than the spacing of this ladder.

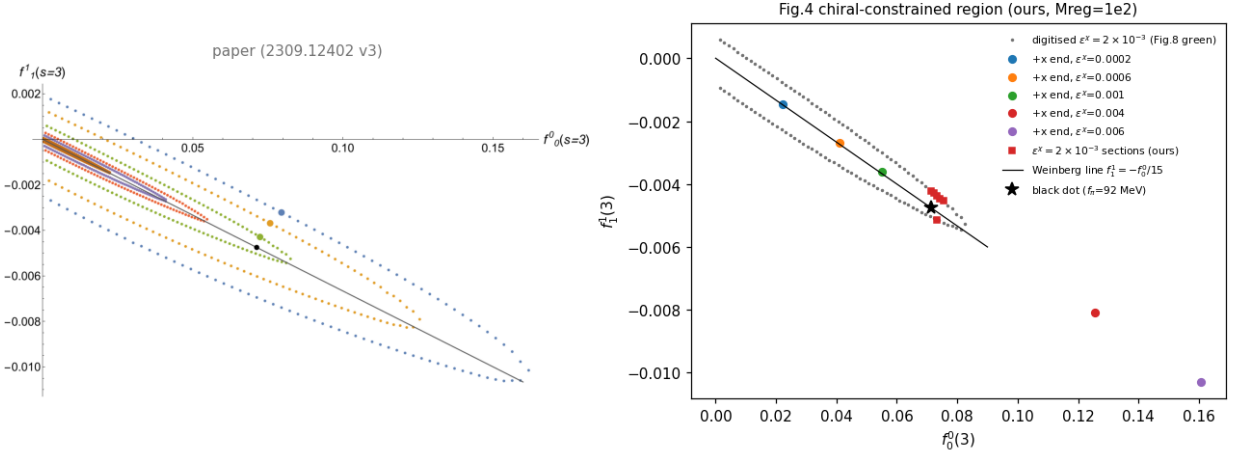


Fig. 4: the paper (left); our six $+x$ ends and the $\epsilon^\chi = 2 \times 10^{-3}$ sections, with the digitised $\epsilon^\chi = 2 \times 10^{-3}$ boundary (right).

3.3 Fig. 5, subthreshold partial waves

“For the larger tolerances the partial waves are not approximately linear ... the chiral zero of the amplitude disappears for the blue points. ... The value $\epsilon^\chi = 0.002$... allows the physical value of f_π .”

The subthreshold curves are evaluated from the accepted solutions at the upper-branch x_{ref} point of each tolerance. They deviate from your digitised curves by at most 6.4 % of $f_0^0(3)$ (RMS), and the S0 zero sits at $s = 0.426$ for 2×10^{-3} , at 0.293 for 4×10^{-3} , and disappears at 6×10^{-3} , as in the figure.

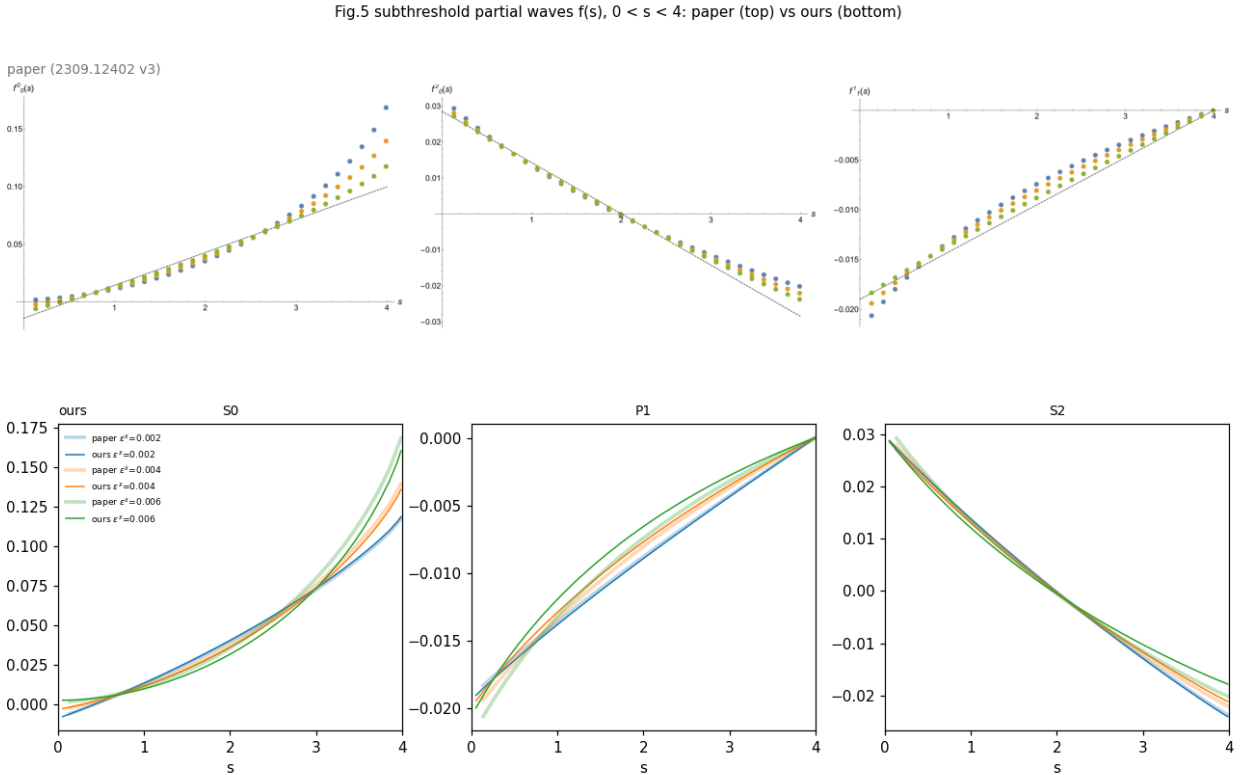


Fig. 5: the three panels of the paper (top) and ours (bottom).

3.4 Fig. 7, chiral-only phase shifts

“We choose a point closest to the black dot ... The partial waves at the magenta point and other nearby agree very well with experimental values for the S0 and S2 waves but not for the P1.”

For the representative point we took, among the upper-branch endpoints of the vertical sections at $x_{\text{ref}} + \{-0.002, -0.001, 0, +0.001, +0.002\}$, the one nearest to the black dot (the rule was fixed before we looked at any phase shift). The RMS deviations from your digitised curves are S0 0.388°, S2 0.144° and P1 0.456°, and P1 shows no 90° crossing below 1.2 GeV.

Fig.7 chiral-only phase shifts at the representative near the black dot: paper (top) vs ours (bottom)

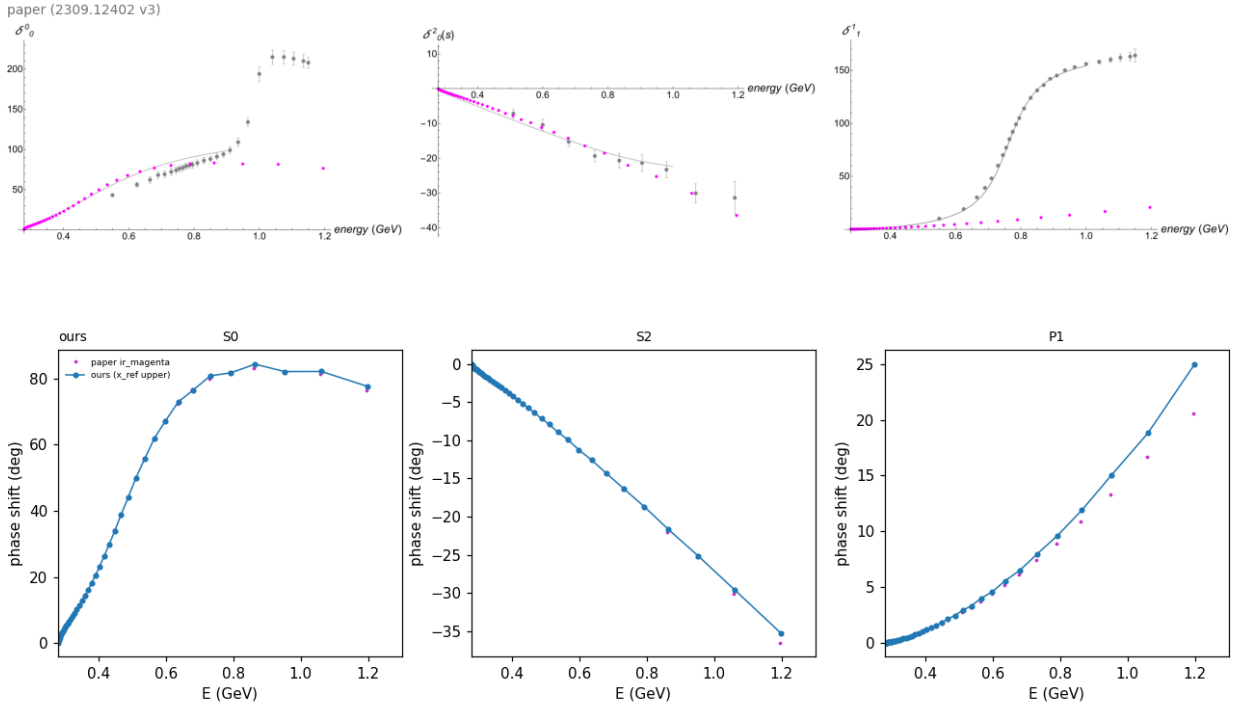


Fig. 7: the paper (top) and ours (bottom).

3.5 Fig. 8, the region after the sum rules and the form-factor bounds

“the plots are produced by taking $\epsilon^{SR} = 2 \times 10^{-3}$ in (3.73) and $\epsilon^{FF} = 6 \times 10^{-5}$ ”; caption: “The upper boundary shrinks notable but the lower not so much.”

We tried three readings of ϵ^{SR} and two of the (3.75) factor; the table gives the values at the x_{ref} section, with your values read off the digitised figure.

Reading	UV +x end	upper shrink	lower rise	ratio
raw box 2×10^{-3} per moment, factor at each node	0.076013	2.8×10^{-4}	2.8×10^{-4}	1
raw box, factor frozen at s_0	0.079623	5.3×10^{-5}	2.5×10^{-4}	0.214
per-wave L^2 ball 2×10^{-3} , factor frozen at s_0	0.079541	5.3×10^{-5}	2.5×10^{-4}	0.214
paper (digitised)	0.08112	2.32×10^{-4}	3.7×10^{-5}	6.3

The relative reading (10% of each moment) is infeasible in this model: dropping the box on the S0

$n = 0$ moment, keeping the other three 10 % boxes and everything else, and minimising that moment gives 3.745×10^{-3} , 1.57 times the QCD value, so nothing in the feasible set gets within 10 % of it. With the raw boxes the solutions sit at the upper edge of three of the four (S0 $n = 0$ at +84 % of its target, S0 $n = 1$ at +1.8 %, P1 $n = 0$ at +1.4 %): the sum rules do bind, and they push towards larger spectral integrals.

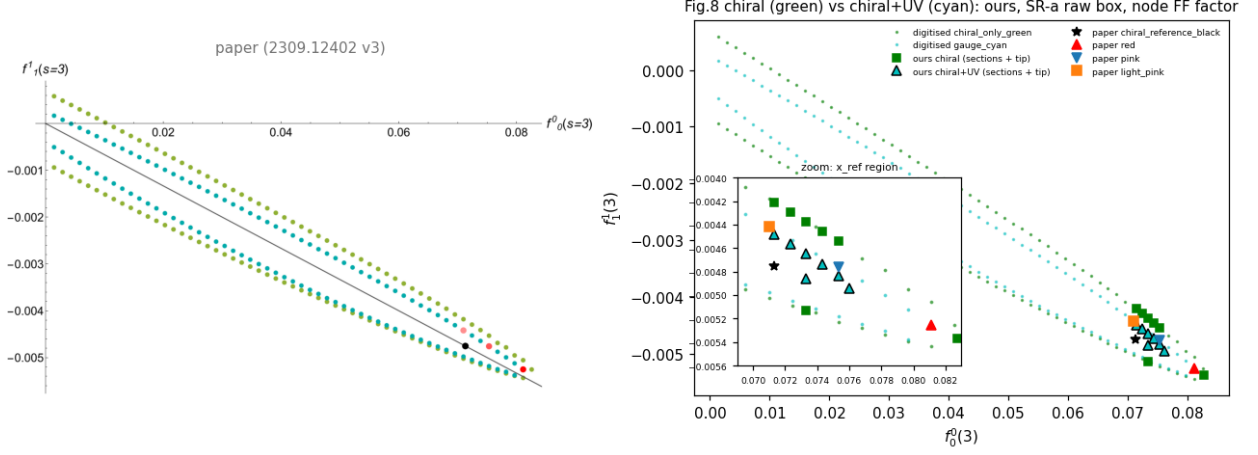


Fig. 8: the paper (left); our chiral and chiral+UV boundary points with the digitised boundaries, and a zoom on the x_{ref} region (right).

3.6 Fig. 9, P1 phase shift

“The resonance energy where the phase shift crosses $\pi/2$ is slightly shifted from the real world data on the mass of the rho at 770 MeV by roughly 6 %.” The digitised crossings are at 0.827, 0.824 and 0.813 GeV.

We use the same three points as before (tip = $+x$ end; reference = the upper-branch point nearest the black dot; mid = its neighbour towards the tip). At the tip the phase crosses 90° at 772.7 MeV with $|S_{P1}| = 0.585$ at 0.79 GeV; at the mid point it crosses at 738.9 MeV with $|S_{P1}|$ dropping to 0.077; at the reference point $|S_{P1}|$ falls to 0.24 at 0.73 GeV and the node values admit two phase branches, the nearest-node lift swinging to -47° while the $+180^\circ$ branch crosses 90° at 708 MeV. With the chiral constraints alone (Fig. 7) there is no crossing below 1.2 GeV, so the resonance does come from the sum rules and the form-factor bounds, as in the paper. It sits 40–110 MeV below yours, and the wave is far from elastic there; the paper does not show $|S|$, so we add it in the second panel.

Fig.9 P1 phase shift after the QCD sum rules (ours: phase and inelasticity)

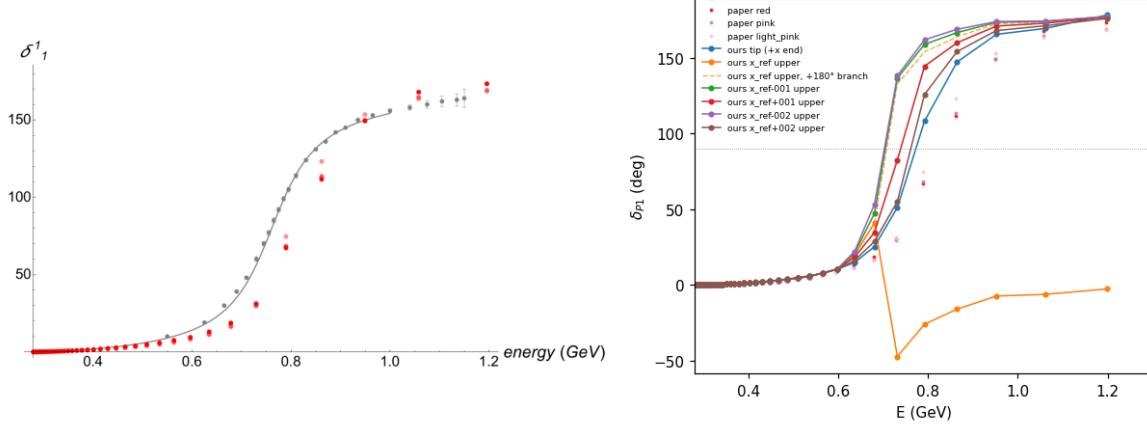
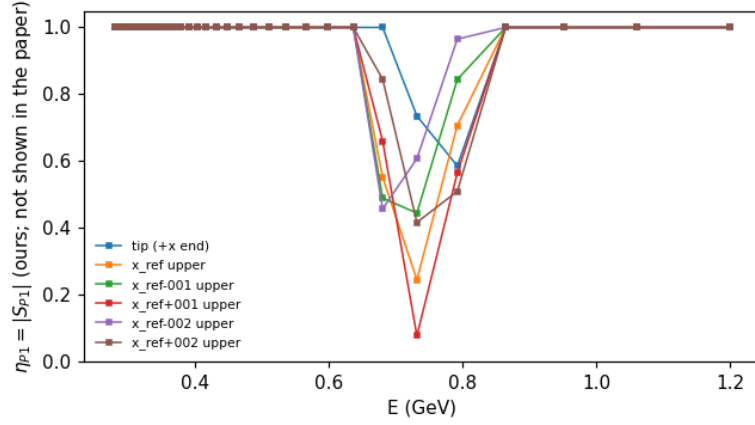


Fig. 9: the paper (left) and our P1 phases at the representatives, with the three digitised curves (right).



$|S_{P1}|$ at the same points (not shown in the paper).

3.7 Fig. 10, S0 and S2 phase shifts

“the three bootstrap (red, pink, light pink) curves ... coincide at low energy but they spread at high energy.”

S2 agrees with your curves (RMS 0.354–1.44°, endpoints –28 to –32° at 1.196 GeV against your –18 to –36°). S0 agrees below about 0.6 GeV and then rises faster than yours, crossing 90° at 0.69 GeV and reaching 181–192° at 1.196 GeV where your curves are at 87–106°. With the chiral constraints alone our S0 matches your Fig. 7 to 0.4°, so this is something the UV constraints do.

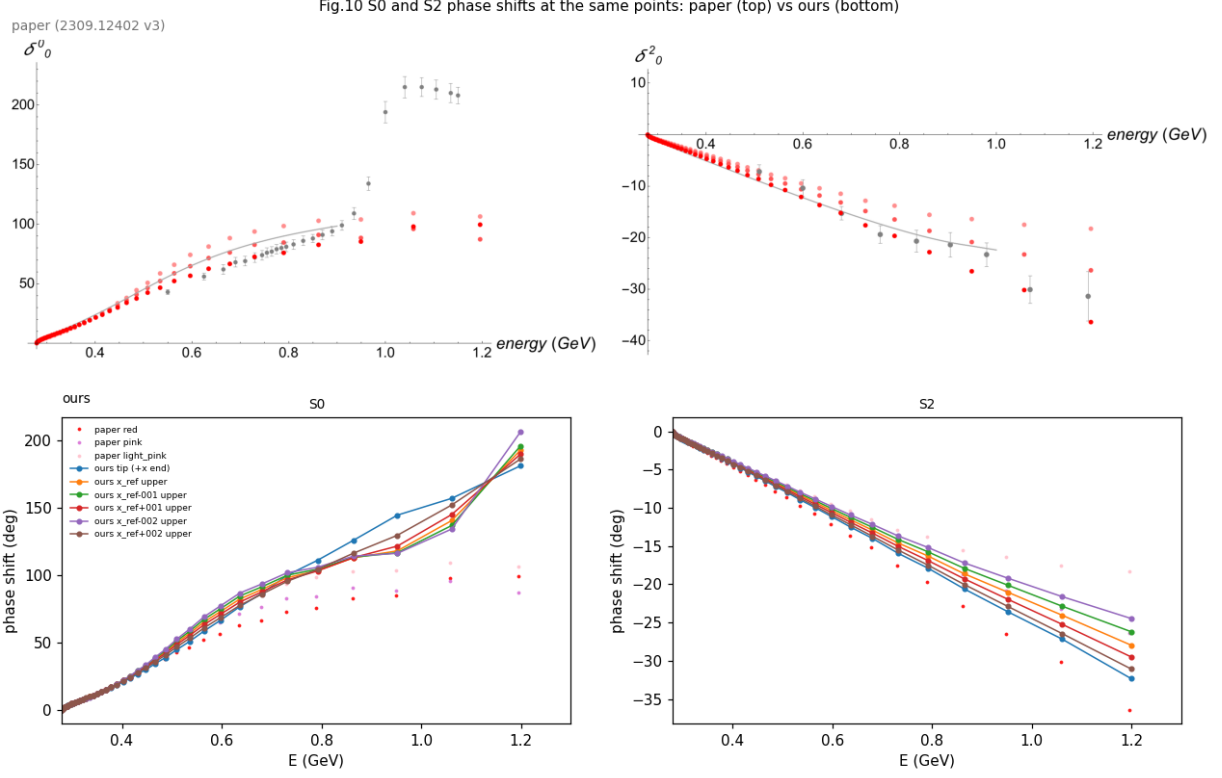


Fig. 10: the paper (top) and ours (bottom).

3.8 Fig. 11, dependence on M and L

“For $M = 50$, we have plotted the results with $L = 8, 10, 12$. In all three partial waves there is a reasonably good convergence. ... the $P1$ phase shifts show a ρ resonance with the peak slightly shifted depending on M .”

(M, L)	UV $+x$ end	P1 crossing (MeV)	$\min S_{P1} $ below 1.2 GeV	S0 at 1 GeV ($^\circ$)
(50, 8)	0.076176	771.4	0.584	151
(50, 10)	0.076013	772.7	0.585	150
(50, 12)	0.075944	772.9	0.585	150
(45, 10)	0.074275	736.1	0.226	165
(60, 10)	0.075097	738.8	0.597	156

In L the crossing moves by 1.5 MeV and the endpoint by 0.3%. In M it moves by 37 MeV, with both $M = 45$ and $M = 60$ below $M = 50$ (in your figure the order is $M = 60 < M = 45 < M = 50$). The S0 phase at 1 GeV is 150–165° for every configuration, against about 95° in yours.

Fig.11 dependence on M and L at the UV tip: paper (top) vs ours (bottom)

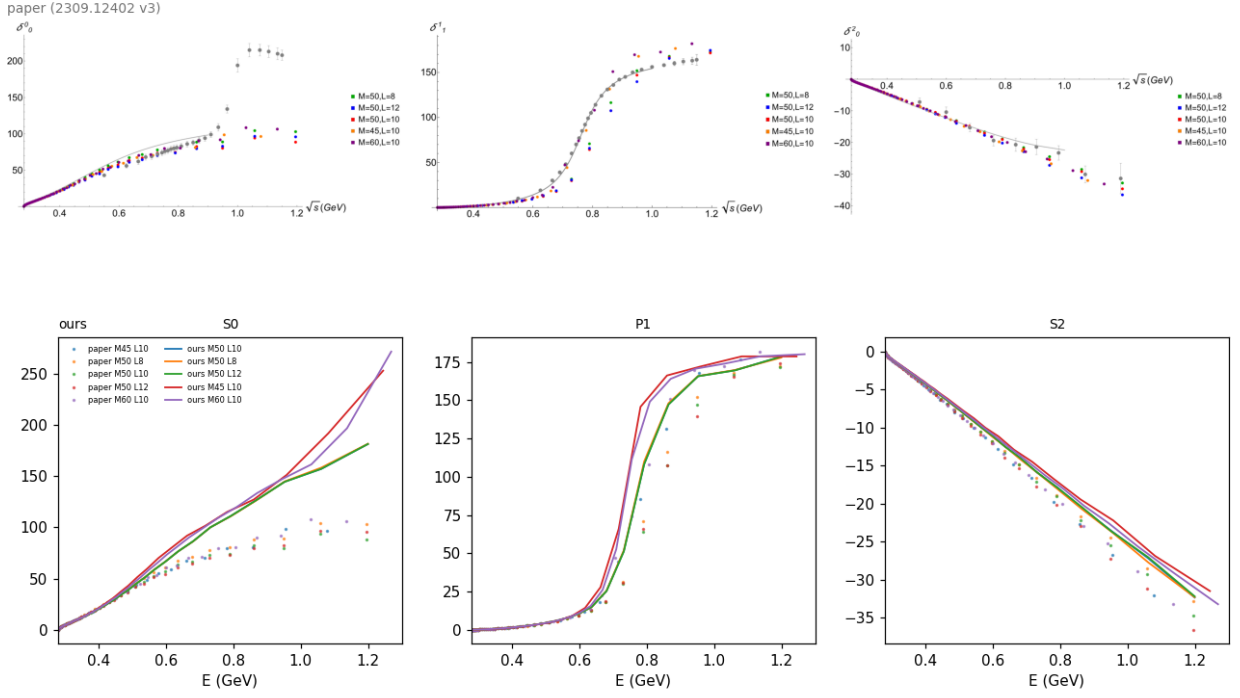


Fig. 11: the paper (top) and ours (bottom).

4 Why the fine features of Figs. 8–10 do not follow from the stated problem

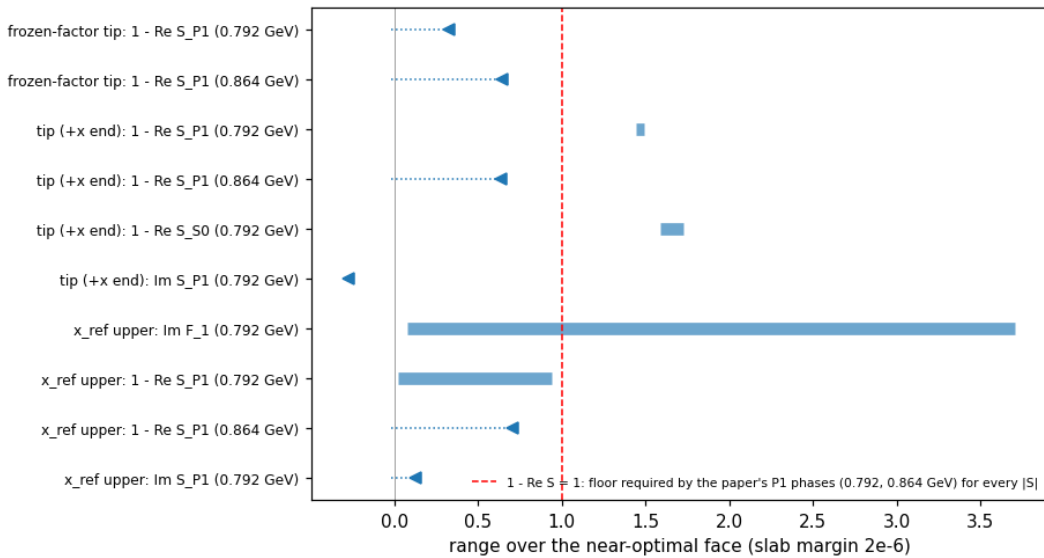
Two more checks on the UV solutions, both exact up to solver tolerance.

Near-optimal face. At a boundary point with support direction d and optimal value v^* we add the constraint $d \cdot (f_0^0, f_1^1) \geq v^* - 2 \times 10^{-6}$ (twice the duality-gap threshold), keep the section, and maximise or minimise one node functional that is linear in the primal variables: $1 - \text{Re } S(s_i)$, $\text{Im } S(s_i)$ or $\text{Im } F(s_i)$. The range of the functional over this slab tells us how much of the amplitude at that boundary point is actually fixed by the extremal problem. Your Fig. 9 phases at the nodes 0.792 GeV ($68.7\text{--}76.2^\circ$) and 0.864 GeV ($112.8\text{--}124.1^\circ$) have $\cos 2\delta < 0$, so any amplitude with those phases has $1 - \text{Re } S > 1$, whatever $|S|$ is.

Representative	Functional	Range over the face	Consequence
tip (+x end)	$1 - \text{Re } S_{P1}$ (0.792 GeV)	[1.441, 1.493]	pinned to ± 0.03 around the tip's own value 1.470, below the 1.72–1.85 of your curves at $ S = 1$
tip	$1 - \text{Re } S_{P1}$ (0.864 GeV)	≤ 0.6346	below 1: your 112.8–124.1° is excluded for every $ S $
tip	$\text{Im } S_{P1}$ (0.792 GeV)	≤ -0.276	negative on the whole face: every amplitude there is already past the resonance at 0.79 GeV
x_{ref} upper (reference)	$1 - \text{Re } S_{P1}$ (0.792 GeV)	[0.02279, 0.9356]	wide, hence undetermined, yet below 1 everywhere: your 68.7–76.2° is excluded for every $ S $
x_{ref} upper	$1 - \text{Re } S_{P1}$ (0.864 GeV)	≤ 0.701	below 1: excluded for every $ S $
x_{ref} upper	$\text{Im } F_1$ (0.792 GeV)	[0.07843, 3.71]	the vector form factor is essentially free at this point
tip	$1 - \text{Re } S_{S0}$ (0.792 GeV)	[1.59, 1.726]	the S0 phase is fixed at 111° on the face and only $ S $ varies (0.79–0.99)

The tip face is rigid in the P1 observables and does not contain your P1 shape. The x_{ref} face is degenerate (a boundary point there does not carry a single set of partial waves) but still excludes your P1 phase at both nodes. So picking a different point on these faces, or a different solver, would not get Fig. 9 out of this constraint set.

Regulariser scale. Raising M_{reg} from 10^2 to 10^3 moves the UV +x end by +0.9%, the P1 crossing by −3.6 MeV and $\min |S_{P1}|$ by −0.025; the Fig. 8 asymmetry does not appear at either scale.



Ranges of the node functionals over the near-optimal faces: bars give [min, max], triangles the maximum where only that was computed; the dashed line is the floor $1 - \text{Re } S = 1$ that the phases of the paper require.

5 After the unitarity-saturation iteration

“One thing we noticed, as you are saying, is that unitarity tends to be unsaturated near the resonance. The rho meson decays primarily to two pions so we used the iterations to correct that. . . . Our results are always shown after that.” (your message of 17 September)

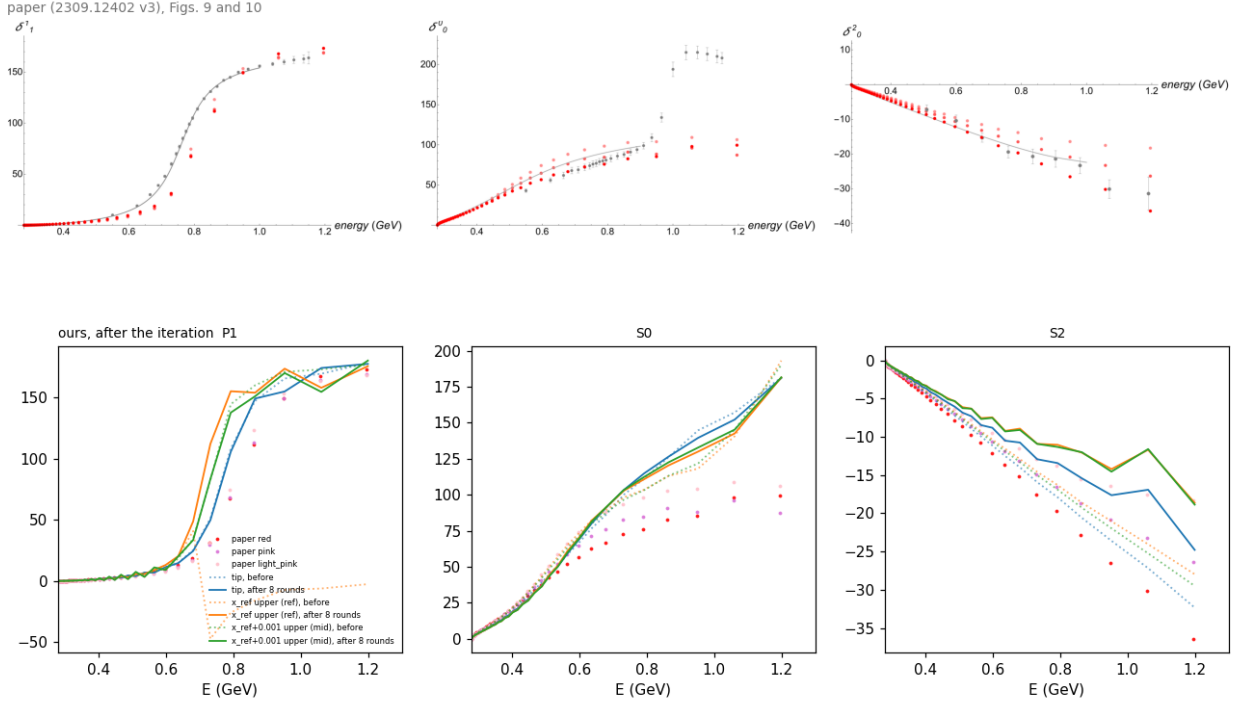
We implemented the step in the form of eq. (2.29) of the follow-up paper: every constraint of the finite problem is kept, the section $f_0^0(3) = x$ is released, and the objective is replaced by $\sum_k \text{Re}[e^{-2i\alpha_k}(S_k - 1)]$ over the nodes below s_0 of S0, P1 and S2, with α_k the phase of the previous form factor (the previous phase shift for S2); each round is a full SDPB solve with the same verification as before. We ran it from the three representatives of Fig. 9, for eight rounds each, and as a control with the point held at its boundary position.

Table 4: Saturation chains, last accepted round (round 0 is the representative of Fig. 9 with the values of Section 3).

Start	Rounds	(f_0^0, f_1^1) at the end	$\min S_{P1} $	ρ (MeV)	S0 at 1 GeV	S2 at 1.2 GeV
tip	8	(0.0678, −0.00449)	0.998	775	145°	−24.8°
x_{ref} upper	8	(0.0632, −0.00419)	0.988	714	135°	−18.6°
$x_{\text{ref}} + 0.001$ upper	8	(0.0635, −0.00422)	0.899	739	138°	−18.9°

The iteration does what it is meant to do: $|S|$ reaches 0.90–1.00 at every node below s_0 and the phases of F and S line up. It does so by leaving the boundary point – the amplitudes drift inwards by 10–15 % in f_0^0 over eight rounds and had not stopped – and when the point is held fixed instead (section kept, near-optimal slab added) the same objective saturates very little: $\min |S_{P1}|$ goes 0.585 $\rightarrow$ 0.593 at the tip and 0.244 $\rightarrow$ 0.493 at x_{ref} . What the iteration does not change is the rest of the picture: the ρ crossings stay at 775, 714, 739 MeV (your 813–827), the S0 wave stays fast, and the three chains do not approach one amplitude. Using your node weights $1/\Lambda_\ell^2$ instead of unit weights changes the drift, not the ρ position (714 against 713 MeV from the same start). The S2 wave, which agreed with yours before, moves away from your curves during the iteration.

After the unitarity-saturation (Watson) iteration: paper (top) vs the converged amplitudes (bottom)



Figs. 9 and 10 of the paper (top) and our amplitudes before (dotted) and after (solid) the saturation iteration (bottom).

6 Sensitivity to the form-factor cap

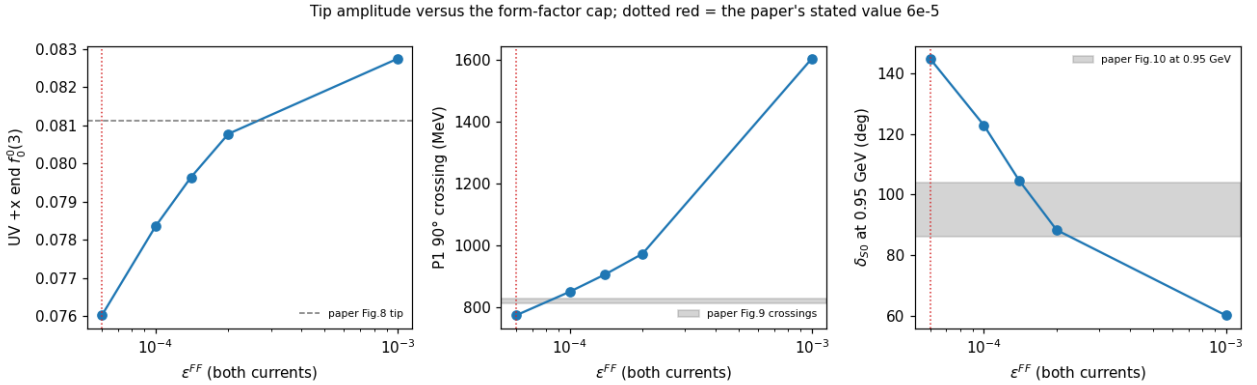
“The faster (or sometimes slower) rise of $S0$ happens, I believe, depending on the parameters. Also changes in the ρ mass.” (same message)

The only continuous parameters of the UV stage are ϵ^{SR} and ϵ^{FF} . ϵ^{SR} turns out not to matter: loosening the raw box from 2×10^{-3} to 10^{-2} , or dropping the $S0$ $n = 0$ box altogether, moves the ρ by at most 12 MeV and leaves $S0$ unchanged (the freed $S0$ $n = 0$ moment settles near 1.9 times the QCD value on its own). ϵ^{FF} in (3.75) matters a great deal, and the two currents act independently.

Table 5: Tip of the UV region against the form-factor cap (factor at each node, everything else as in the main line).

ϵ^{FF}	$+x$ end	ρ (MeV)	$\min S_{P1} $	δ_{S0} at 0.79/0.86/0.95/1.06 GeV
6×10^{-5} (both currents)	0.076013	772.7	0.585	111/126/145/157
1×10^{-4} (both currents)	0.078351	849.1	0.785	92.5/107/123/146
1.4×10^{-4} (both currents)	0.079637	905.3	0.845	81.8/91.9/105/130
2×10^{-4} (both currents)	0.080779	972.3	0.973	73/81.7/88.3/111
1×10^{-3} (both currents)	0.082753	1604	0.987	59.5/59.7/60.3/57.7
S0 cap 2e-4, P1 cap 6e-5	0.077579	775.3	0.585	74.9/84/90.1/111
S0 cap 6e-5, P1 cap 2e-4	0.07891	966.6	0.956	110/125/148/164
paper	0.0811	813–827	–	76/83/86/98 (red), 99/103/104/109 (light pink)

With your stated 6×10^{-5} the S0 wave is too fast and the ρ too low. Loosening the S0 cap alone to 2×10^{-4} puts S0 on your red curve to within 3° up to 0.95 GeV without touching the ρ ; loosening the P1 cap alone moves the ρ up (to about 850 MeV at 10^{-4} and 970 at 2×10^{-4}) without touching S0. Your figures therefore correspond to an effective constraint on the form factors above s_0 that is looser than our reading of (3.75) with 6×10^{-5} , and looser for the scalar current than for the vector one; the same change carries the $+x$ end of Fig. 8 from our 0.0760 towards your 0.0811. We have not returned anything on the strength of this – the table is a scan – but it does say where the remaining difference sits.



Tip amplitude against ϵ^{FF} (both currents); the grey bands are the paper's values, the dotted line its stated 6×10^{-5} .

7 A request

It would help us a great deal to see the code or notebooks behind the 2309 runs, in particular the parts that implement (3.73) and (3.75) – how the caps on $\mathcal{F}_0$ and $\mathcal{F}_1$ above s_0 were normalised – and the saturation iteration with the way the three points of Figs. 9 and 10 were selected. With that we could tell which of the readings above you used and settle the remaining differences; we are happy to share any of our solutions and the full log of our runs in return.

References

- Y. He and M. Kruczenski, *Bootstrapping gauge theories*, arXiv:2309.12402 (the paper reproduced here).
- Y. He and M. Kruczenski, *Gauge Theory Bootstrap: Pion amplitudes and low energy parameters*, arXiv:2403.10772 (the follow-up paper; its released code was read, not run).
- Y. He and M. Kruczenski, *The Gauge Theory Bootstrap: Predicting pion dynamics from QCD*, arXiv:2505.19332 (current code of the repository `gauge-theory-bootstrap`).
- Y. He and M. Kruczenski, *S-matrix bootstrap in 3+1 dimensions: regularization and dual convex problem*, JHEP 08 (2021) 125, arXiv:2103.11484 (the earlier method paper whose Section 3 introduces the regularisation of the double spectral density).